
Assignment 2

C Programming — If-Else & Nested Conditions

Question 1: Greatest of 4 Numbers using Nested If-Else

```
// Write a program to find the greatest number among 4 numbers using nested if-else

#include <stdio.h>

void main() {
    int a, b, c, d;
    printf("Enter four numbers: ");
    scanf("%d %d %d %d", &a, &b, &c, &d);

    if (a > b) {
        if (a > c) {
            if (a > d)
                printf("%d is greatest\n", a);
            else
                printf("%d is greatest\n", d);
        } else {
            if (c > d)
                printf("%d is greatest\n", c);
            else
                printf("%d is greatest\n", d);
        }
    } else {
        if (b > c) {
            if (b > d)
                printf("%d is greatest\n", b);
            else
                printf("%d is greatest\n", d);
        } else {
            if (c > d)
                printf("%d is greatest\n", c);
            else
                printf("%d is greatest\n", d);
        }
    }
}
```

Question 2: Check Leap Year

```
// Write a program to check a given year is a leap year or not
#include <stdio.h>
```

```
void main() {
    int year;
    printf("Enter year: ");
    scanf("%d", &year);

    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
        printf("%d is a leap year\n", year);
    else
        printf("%d is not a leap year\n", year);
}
```

Question 3: Check Positive, Negative or Neutral

```
// Write a program to check if a given number is positive, negative or neutral

#include <stdio.h>

void main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);

    if (num > 0)
        printf("%d is positive\n", num);
    else if (num < 0)
        printf("%d is negative\n", num);
    else
        printf("Number is neutral (zero)\n");
}
```

Question 4: Check Palindrome (3-Digit Number)

```
// Write a program to check if a given 3-digit number is a palindrome or not

#include <stdio.h>

void main() {
    int num, first, second, third, rev;
    printf("Enter a 3-digit number: ");
    scanf("%d", &num);

    first = num / 100;
    second = (num / 10) % 10;
    third = num % 10;

    rev = (third * 100) + (second * 10) + first;

    if (num == rev)
        printf("%d is a palindrome number\n", num);
    else
        printf("%d is not a palindrome number\n", num);
}
```

Question 5: Check Armstrong Number

// Write a program to check if a given 5-digit number is an Armstrong number or not

```
#include <stdio.h>

void main() {
    int num, temp, d1, d2, d3, sum;
    printf("Enter a 3-digit number: ");
    scanf("%d", &num);

    temp = num;
    d1 = temp % 10;
    temp = temp / 10;
    d2 = temp % 10;
    temp = temp / 10;
    d3 = temp % 10;

    sum = (d1 * d1 * d1) + (d2 * d2 * d2) + (d3 * d3 * d3);

    if (sum == num)
        printf("%d is an Armstrong number\n", num);
    else
        printf("%d is not an Armstrong number\n", num);
}
```

Question 6: Check Uppercase or Lowercase Character

// Write a program to check whether a character is uppercase or lowercase

```
#include <stdio.h>

void main() {
    char ch;
    printf("Enter a character: ");
    scanf("%c", &ch);

    if (ch >= 'A' && ch <= 'Z')
        printf("%c is uppercase\n", ch);
    else if (ch >= 'a' && ch <= 'z')
        printf("%c is lowercase\n", ch);
    else
        printf("%c is not an alphabet\n", ch);
}
```

Question 7: Check Vowel or Consonant

// Write a program to check whether a character is a vowel or a consonant

```
#include <stdio.h>
```

```
void main() {
    char ch;
    printf("Enter a character: ");
    scanf("%c", &ch);

    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
        ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U')
        printf("%c is a vowel\n", ch);
    else
        printf("%c is a consonant\n", ch);
}
```

Question 8: Check Divisibility by 5 and 11

```
// Write a program to check whether a number is divisible by 5 and 11

#include <stdio.h>

void main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);

    if (n % 5 == 0 && n % 11 == 0)
        printf("%d is divisible by both 5 and 11\n", n);
    else
        printf("%d is not a multiple of 5 and 11\n", n);
}
```

Question 9: Check if Number is Between 100 and 200

```
// Program to check whether a number lies between 100 and 200 or not

#include <stdio.h>

void main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);

    if (n >= 100 && n <= 200)
        printf("%d is inside 100 and 200 range\n", n);
    else
        printf("%d is outside 100 and 200 range\n", n);
}
```

Question 10: Calculate Electricity Bill

```
// Calculate electricity bill according to units consumed

#include <stdio.h>
```

```
void main() {
    int units;
    float bill;
    printf("Enter units consumed: ");
    scanf("%d", &units);

    if (units <= 100)
        bill = units * 1.5;
    else if (units <= 200)
        bill = 100 * 1.5 + (units - 100) * 2.5;
    else
        bill = 100 * 1.5 + 100 * 2.5 + (units - 200) * 3.5;

    printf("Electricity bill = %.2f\n", bill);
}
```